Predictive Models for National Park Activity Fees

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GitHub Repository: https://github.com/lareina1515/NPS_Predictive_Modeling

1. Introduction

National parks offer a wide range of recreational and educational activities, and establishing appropriate fee policies is an important consideration in park management. Park managers may need to balance funding and operating costs with public access and the fairness of charging for recreational services. Understanding how fee status varies across different activities could provide useful context for reviewing existing fee arrangements and planning new activities. This motivates our project to investigate how activity characteristics influence pricing and provide park managers with data-driven evidence to inform decisions about whether fees should be charged for specific activities.

Previous research has approached recreational fee policies from several perspectives. Schwartz (2005) highlighted the importance of considering fee collection costs alongside the revenue generated by fees at U.S. national parks. Public access is another important concern: Ostergren et al. (2005) explored visitors' views on NPS fees, including their preferences for fee structures and concerns about potential barriers to visitation. Chung et al. (2011) further examined how perceptions of price fairness relate to visitors' willingness to pay for recreational services. Taken together, these studies suggest that fee policies involve not only financial and operational considerations but also questions of public access and fairness.

Although previous studies have examined fee collection costs, public attitudes, and willingness to pay, less attention has been given to predicting the fee status of individual park activities based on their characteristics. To address this gap, our project uses publicly available data from the NPS to investigate whether features such as activity categories, reservation requirements, topics, seasons, and times of day can help predict whether a fee applies to a specific activity.

2. Data

2.1 Data Sources & Data Collection

For this project, we collected the raw dataset from the U.S. NPS API (*API Documentation - Developer Resources (U.S. National Park Service)*, n.d.). We sent requests to the API to retrieve information on suggested visitor activities (thingstodo) that are recommended by individual national parks and protected sites.

In filtering the raw dataset through column selection, we dropped numerous high-dimensional and redundant features to focus strictly on our analytical goals. For example, unstructured text fields such as `longDescription` were excluded because they fall outside the scope of our tabular analysis. We also removed several attributes with severe missing values or limited predictive values for our modeling goals (e.g., `amenities` and `url`).

To prepare the dataset for exploratory data analysis, we further transformed several remaining features. Specifically, the original `tags` and `topics` fields contained hundreds of different keywords, which we converted into integer counts to quantify the diversity and richness of each activity. From the `relatedParks` column, we extracted state information to define geographic categories based on U.S. Census Bureau regions and additional territories (Bureau, n.d.). Furthermore, we mapped a wide variety of park designations into five main park categories according to their primary purpose, such as nature conservation, historical preservation. Informed by the NPS descriptions of park types (*What's In a Name?*, n.d.), we developed this classification specifically for our analysis. By pulling the necessary details from `relatedParks` and removing redundant fields, we finalized a clean dataset ready for subsequent modeling.

2.2 Dataset Properties & Key Features

The final dataset used for model building and analysis contains 3,571 samples and 9 fields: 3 boolean variables (`doFeesApply`, `isReservationRequired`, `arePetsPermitted`) with values of either True or False; 2 discrete count numerical variables (`tags`, `topics`) representing activity diversity; and 4 multi-label categorical variables (`timeOfDay`, `season`, `regions`, `park_categories`) recording one or more corresponding categories in the form of lists. Among these attributes, `doFeesApply` serves as the primary binary target variable, indicating whether an activity requires additional entry or participation fees. Key predictors such as `isReservationRequired` capture booking restrictions, while features like `timeOfDay` and `season` provide temporal context to help predict activity fee structures.

3. Exploratory Data Analysis

After cleaning the dataset, our team carried out Exploratory Data Analysis (EDA) to get a better understanding of the data and identify patterns and relationships between variables. We used several visualizations to explore the distributions of the features and their relationships with the target variable, which helped guide our feature selection and modeling decisions.

The pie chart on the left shows that 80% of activities do not require a fee, while only 20% do, indicating that the target variable is imbalanced. Considering that a model predicting "no fee" for every activity would achieve 80% accuracy, it would fail to identify any fee-requiring activities. In that case, multiple evaluation metrics should be applied to monitor the overall performance of the model.

The charts (center and right) also show that only 9.3% of activities require reservations, while 54.2% allow pets. These features may help distinguish different types of activities, although further analysis is needed to determine how strongly they are related to fee requirements.

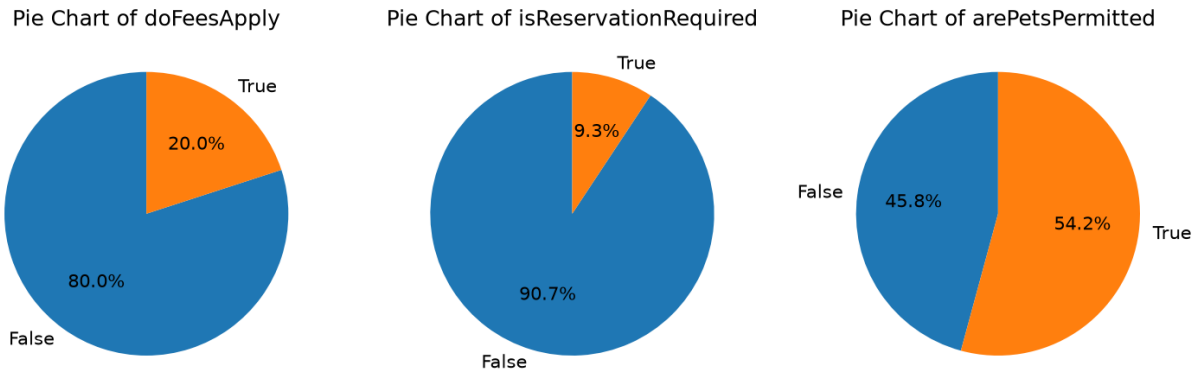


Fig 1. Pie charts showing the distributions of binary variables indicating whether the activity requires fees, reservations, and whether pets are allowed.

The violin plot shows that most activities have a relatively low number of tags. The average is 5.49 for activities without fees and 5.72 for activities with fees. However, the median is slightly higher for activities without fees (5 vs. 4.5). The middle 50% of activities have 3 to 8 tags for the no-fee group and 3 to 7 tags for the fee group.

Overall, the two groups have similar tag counts, although activities without fees have a slightly wider distribution at higher values. This suggests that tag count alone may not be a strong predictor of fee requirements, but it could still be useful when combined with other features. The unusually high tag counts are also worth checking to see whether they are valid values.

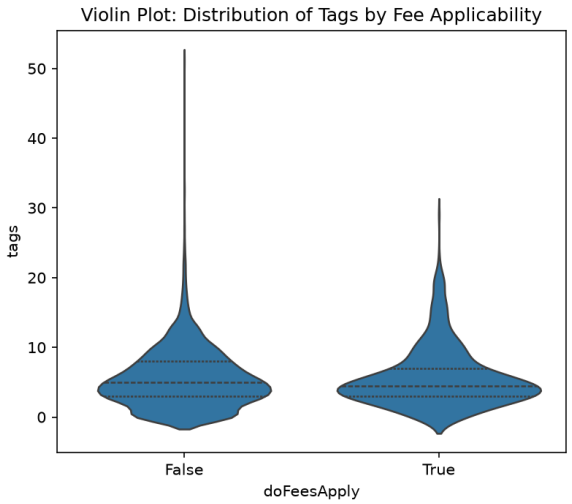


Fig 2. Violin plot comparing the distributions of tags for activities with and without fees.

The heatmap demonstrates that most activities are concentrated in the lower ranges of both variables; the majority of co-occurrence happens between topics with counts 0-7 and tags with counts 0-13. The highest frequencies appear among activities with relatively few tags and topics, while combinations with larger counts are less common.

This pattern suggests that most activities have concise metadata descriptions. Tags and topics may also be related to each other, so it would be useful to examine their correlation and compare how each variable relates to the target, helping to determine whether they provide useful information to the model.

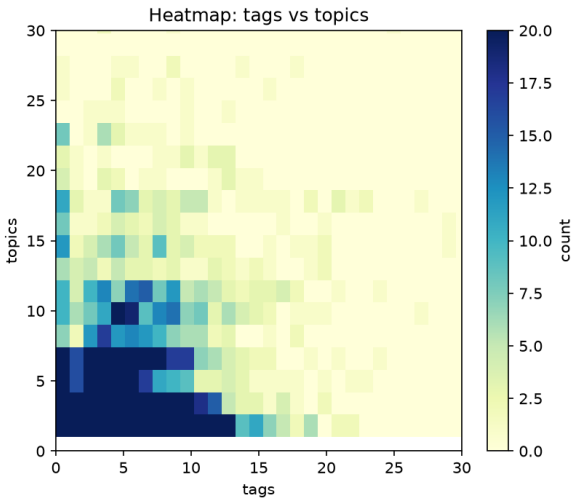


Fig 3. Heatmap showing the joint frequency of tags and topics, with darker cells representing more observations.

The stacked bar charts show that national parks and historical sites contain the largest numbers of activities. Geographically, the West and South account for the most observations. Fee-requiring

activities appear across all the displayed categories and regions. In addition, clear differences in fee rates across park categories and regions are also important. Recreation areas have the highest percentage of fee-requiring activities (27.1%), while national monuments have the lowest (16.9%). By region, the South has the highest fee rate (25.6%), whereas the Northeast has the lowest (12.2%).

The differences suggest that both park category and geographic region may help predict whether an activity requires a fee, making them potentially useful features for the classification model.

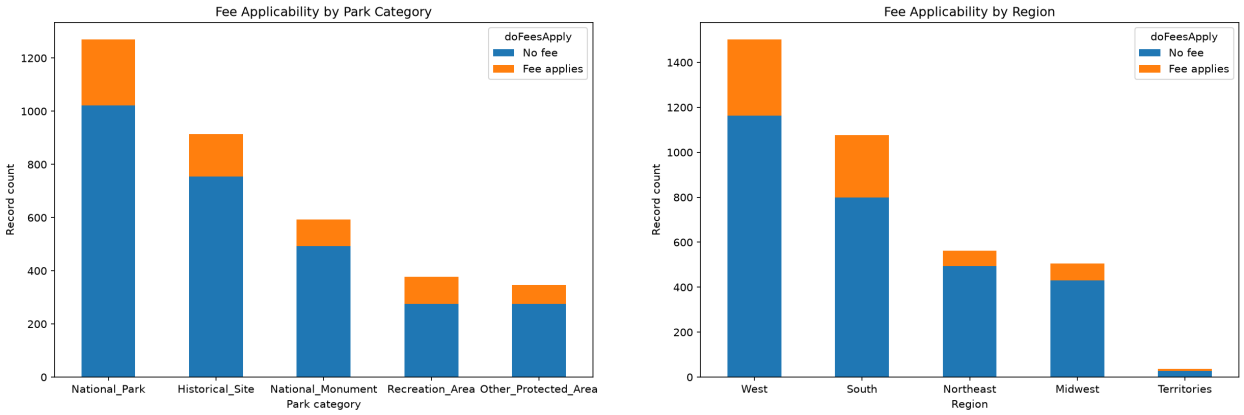


Fig 4. Stacked bar charts showing counts of fee-free and fee-requiring activities by park category and region.

4. Problem Description and Potential Impact

We formulate our project as a binary classification problem. The target variable, `doFeesApply`, indicates whether a listed activity requires a fee. The model classifies activities into two categories: those for which a fee applies and those for which no fee applies.

We aim to predict activity fee status based on available activity characteristics, including reservation requirements, pet policies, tags, topics, seasons, and times of day. By analyzing the relationships between these features and existing fee requirements, our project seeks to identify patterns that may help explain differences in fee status across activities. Categorical and non-numerical features will be appropriately processed to support the analysis.

The potential value of our project is to provide park managers with data-driven evidence to support activity-level fee decisions. By identifying characteristics associated with fee status, our analysis may help managers understand existing fee patterns and identify activities that warrant further review. Comparing similar activities with different fee statuses could also help managers evaluate current fee arrangements and plan future activities. However, fee status alone cannot determine whether charging for an activity is appropriate. Final decisions should also consider

factors such as operating costs, park regulations, management objectives, and equitable public access, which may not be fully captured in the dataset. Thus, our model is intended to inform, rather than replace, managerial judgment.

5. References

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